

Lecture - 6

Matrix in Row Echelon form

An $m \times n$ matrix is said to be in reduced row echelon form if it satisfies the following conditions.

- ① The left most nonzero entry in every row is 1. Such an entry is called a **pivot**.

$$\begin{pmatrix} 0 & \boxed{1} & 3 & 0 \\ 0 & 0 & 0 & \boxed{1} \\ \boxed{1} & 0 & 0 & 0 \end{pmatrix} \quad \text{--- Example (a)}$$

- ② The Column containing a pivot must have all other entries 0.

$$\begin{pmatrix} 0 & \boxed{1} & 3 & 0 \\ 0 & 0 & 0 & \boxed{1} \\ \boxed{1} & 0 & 0 & 0 \end{pmatrix} \quad \begin{array}{l} \text{Obey's ① and ②} \\ \text{--- Example (a)} \end{array}$$

- ③ If $i < j$ and $a_{ip} = 1$ is a pivot and $a_{jq} = 1$ is also a pivot then $p < q$

Example (a) doesn't obey this rule

However the following example obeys ①, ② and ③

$$\left(\begin{array}{ccccc} \circ & \circ & \circ & \circ & \circ \\ \circ & | & \circ & \circ & \circ \\ \circ & \circ & | & \circ & \circ \\ \circ & \circ & \circ & \circ & | \end{array} \right) \quad \text{---} \quad \textcircled{b}$$

④ All the Zero rows occur at the bottom

Example ⑤ doesn't obey the rule ④

Examples

(i) $\left(\begin{array}{cccc} \circ & | & \circ & \circ \\ \circ & \circ & | & \circ \\ \circ & \circ & \circ & | \end{array} \right)$

(iii) $\left(\begin{array}{cccccc} \circ & | & \circ & \circ & \circ & \circ \\ \circ & \circ & \circ & | & \circ & \circ \\ \circ & \circ & \circ & \circ & \circ & | \end{array} \right)$

(ii) $\left(\begin{array}{ccccc} \circ & | & \circ & \circ & \circ \\ \circ & \circ & | & \circ & \circ \\ \circ & \circ & \circ & \circ & | \\ \circ & \circ & \circ & \circ & \circ \end{array} \right)$

(iv) $\circ$ matrix

Non examples

$$(i) \begin{pmatrix} 1 & 0 & 2 & 3 \\ 0 & 1 & 2 & 4 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

$$(iii) \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}$$

$$(ii) \begin{pmatrix} 4 & 2 & 3 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

$$(iv) \begin{pmatrix} 0 & 0 & 0 \\ 1 & 2 & 3 \\ 0 & 0 & 0 \end{pmatrix}$$

Row reduced Echelon form of a matrix

Row operations can be used to put a matrix in Row Echelon form.

We saw that

$$\textcircled{1} \begin{pmatrix} 2 & 5 & 2 \\ 3 & 2 & 1 \\ 4 & -1 & 3 \end{pmatrix} \xrightarrow{\text{Row operations}} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$A \xrightarrow{\quad} I_3$
Row reduced echelon form of A is I_3

② We also saw that

$$\left(\begin{array}{ccc|c} 2 & 5 & 2 & 0 \\ 0 & -11 & -1 & -1 \\ 4 & -1 & 3 & -1 \end{array} \right) \longrightarrow \left(\begin{array}{ccc|c} 1 & 0 & \frac{17}{22} & \frac{5}{22} \\ 0 & 1 & \frac{1}{11} & \frac{1}{11} \\ 0 & 0 & 0 & 0 \end{array} \right)$$

Some Facts (without Proofs!)

① Note that performing a row operation is equivalent to multiplying a suitable elementary matrix to the left of the given matrix.

Let E_{ij} be the matrix whose ij^{th} entry is 1 and 0 everywhere

② $R_i \rightarrow \alpha R_i$ is obtained by premultiplying the matrix

$$B = E_{11} + \dots + E_{i-1,i-1} + \alpha E_{ii} + E_{i+1,i+1} + \dots + E_{nn}$$

③ $R_i \leftrightarrow R_j$ is obtained by premultiplying the matrix by say $i < j$

$$B = E_{11} + E_{12} + \dots + E_{i-1,i-1} + \boxed{E_{ij}} + E_{i+1,i+1} + \dots + E_{j-1,j-1} + \boxed{E_{ji}} + E_{j+1,j+1} + \dots + E_{nn}$$

④ $R_i \rightarrow R_i + \alpha R_j$; $i < j$ is obtained by premultiplying the matrix by $B = I_n + \alpha E_{ij}$

(ii) Every matrix can be made into a row echelon matrix by performing row operations! We saw many examples

(iii) Given a matrix A , there is a matrix B such that BA is in row echelon form and that $B = B_1 B_2 \cdots B_l$, where each B_i is an elementary matrix. We saw this as well.

(iv) Row echelon form of a matrix is unique. Proof will be given later

Invertible Matrices

Proposition

Let A be an $n \times n$ matrix. Then the following statements are equivalent

- (a) Reduced row echelon form of A is I_n
- (b) Reduced row echelon form matrix of A has no zero rows

Proof

(a) $\Rightarrow$ (b) is clear

⑥ $\Rightarrow$ ① If the echelon form of A has no zero rows then each row must have 1. Since A is an $n \times n$ matrix, from rule ③, we obtain that $a_{11}=1, a_{22}=1, \dots, a_{nn}=1$

Thus, we obtain that the echelon form is I_n . $\square$